MSc Natural Language Processing – Group Assignment

Contents

NLP Group Assignments (20%) - Groups 1 to 19 1
Overview 1
Evaluation 2
General Workflow 2
Topic 1 – Sentiment Analysis 2
Topic 2 – Machine Translation 3
Topic 3 – Named Entity Recognition (NER) 4
NLP Group Assignments (40%) - Groups 24 & 25 5
Advanced Topic – Multilingual Sentiment Analysis 5
Evaluation 6
Deliverables (in both cases) 6
Technical report (in both cases) 6

NLP Group Assignments (20%) - Groups 1 to 19

Groups 1-19, as established via the Canvas, fall under this category.

Overview

This group assignment aims to provide students with hands-on experience in designing, implementing, and evaluating Natural Language Processing (NLP) systems. Students will work in groups to implement and compare different approaches to solve a real-world NLP task. The assignment emphasizes the full NLP pipeline: data preparation, text representation, model development, evaluation, and analysis. The comparison between Baseline and Neural-based models will empower students with insights as concerns the implementation strategy that needs to be followed in these two different approaches and the pre-processing needs. The latter will help on the identification of advantages and disadvantages and a clear distinction between these two approaches.

Each group will select one of the following three topics:

- Sentiment Analysis

- Machine Translation
- Named Entity Recognition (NER)

In addition, one group will be assigned an advanced task involving Multilingual Sentiment Analysis, which will carry a higher weight in the final grade.

Evaluation

Criterion	**Weight**
Implementation	40%
Experimental design	30%
Results analysis/comparison	20%
Report quality	10%

General Workflow

Regardless of the selected topic, all groups should follow a structured workflow:

1. Problem Definition
2. Dataset Selection and Description
3. Data Preprocessing
4. Text Vector Representation and Embeddings
5. Model Development
6. Model Training and finetuning
7. Evaluation
8. Error Analysis and Discussion

Topic 1 – Sentiment Analysis

Sentiment Analysis is the task of automatically identifying the sentiment expressed in text. The goal is to determine whether a piece of text expresses a positive, negative, or neutral opinion. This task is widely used in applications such as product review analysis, social media monitoring, customer feedback analysis, and market research.

In this assignment, students must build a system that classifies the sentiment of text samples such as movie reviews, product reviews, or social media posts.

Students should begin by selecting a dataset appropriate for sentiment analysis. Common datasets include:

Yelp open dataset (also available in PyTorch,	https://business.yelp.com/data/resources/open-dataset/

torchtext.datasets.yelpreviewfull)	
Amazon reviews dataset (also available in PyTorch, torchtext.datasets.amazonreviewfull)	https://amazon-reviews-2023.github.io/
Sentiment140 dataset	https://www.kaggle.com/datasets/kazanova/sentiment140
Multiclass SA dataset	https://huggingface.co/datasets/Sp1786/multiclass-sentiment-analysis-dataset
Twitter US Airline Sentiment	https://www.kaggle.com/datasets/crowdflower/twitter-airline-sentiment

The preprocessing stage should include steps such as tokenization, lowercasing, handling punctuation, and potentially removing stop words (whenever is deemed necessary and appropriate). Students should also consider how to handle unknown words (i.e., OOV – Out of Vocabulary) and how to manage variable-length sequences.

Students must implement at least two models:

- A baseline model using classical NLP techniques (e.g., TF-IDF + Logistic Regression or Naive Bayes)
- A neural model such as an LSTM, CNN for text classification, or a Transformer-based model like BERT.

The models must be evaluated using metrics such as accuracy, precision, recall, and F1-score. Students should also include a confusion matrix and perform a qualitative analysis of misclassified examples.

Topic 2 – Machine Translation

Machine Translation (MT) is the task of automatically translating text from one language into another. Modern machine translation systems are widely used in applications such as multilingual communication, localization of content, and cross-language information access.

In this assignment, students will build a system that translates text between two languages (for example English–French, English–German, or English–Spanish).

Students should select an appropriate dataset containing parallel sentences. Suitable datasets include:

Multi30K (also available in PyTorch, torchtext.datasets. multi30k)	https://huggingface.co/datasets/bentrevett/multi30k
FameMT	https://github.com/laniqo-public/fame-mt
MT560	https://opus.nlpl.eu/MT560

EuroParl	https://www.statmt.org/europarl/
Tatoeba	https://tatoeba.org/en/downloads
Link with other MT datasets	https://www2.statmt.org/wmt24/mtdata/

The preprocessing stage should include tokenization and possibly subword segmentation techniques such as Byte Pair Encoding (BPE) and SequencePiece. Students should also address sequence length handling and vocabulary construction.

Students must implement:

- A baseline approach (for example a simple sequence-to-sequence encoder–decoder model)
- A neural machine translation model such as a Transformer architecture or a pretrained translation model (e.g., MarianMT).

Evaluation should include BLEU score computation as well as qualitative analysis of translation outputs. Students should present several examples of generated translations and discuss common errors such as incorrect word order, missing words, or semantic inaccuracies.

Topic 3 – Named Entity Recognition (NER)

Named Entity Recognition is a sequence labeling task in which the goal is to identify and classify entities in text into predefined categories such as persons, organizations, locations, or dates.

This task is important in many NLP applications including information extraction, question answering, knowledge graph construction, and document analysis.

Students should use a labeled NER dataset such as the below:

Conll2003	https://huggingface.co/datasets/eriktks/conll2003
MIT Movie Corpus or MIT Restaurant Corpus	https://sls.csail.mit.edu/downloads/
WNUT17	https://huggingface.co/datasets/leondz/wnut_17
WikiGold	https://huggingface.co/datasets/milistu/Wikigold-NER-conll
Additional link (with other languages if you want to experiment)	https://github.com/ay94/NER-datasets

The preprocessing stage should include tokenization and conversion of labels into a tagging scheme such as BIO or BIOES.

Students must implement:

- A baseline model such as Conditional Random Fields (CRF)
- A neural model such as BiLSTM or BiLSTM+CRF or a transformer-based approach such as BERT fine-tuning.

Evaluation must be performed using precision, recall, and F1-score at the entity level. Students should also analyze common errors such as partial entity detection or confusion between entity types.

NLP Group Assignments (40%) - Groups 24 & 25

Groups 24 & 25, as established via the Canvas, fall under this category.

Advanced Topic – Multilingual Sentiment Analysis

These two groups will work on a more advanced task: multilingual sentiment analysis. The objective is to build a multilingual sentiment classifier that can operate across different languages (at least three languages should be examined). The workflow should follow the instructions presented also for the other groups.

This task introduces challenges such as cross-lingual representation, language transfer, and multilingual embeddings.

Students must experiment with multilingual models such as:

- Multilingual FastText embeddings
- Multilingual BERT (mBERT)
- XLM-R or LaBSE or LASER

They should **compare at least three different models** and investigate whether a model trained in one language can generalize to other languages and analyze cross-lingual performance differences.

Datasets that can be used are as per below:

Amazon Reviews Multi	https://www.kaggle.com/datasets/mexwell/amazon-reviews-multi
M-ABSA dataset	https://github.com/swaggy66/M-ABSA
Social media Multilingual SA	https://www.kaggle.com/datasets/suraj520/multi-task-learning
Multilingual SA	https://huggingface.co/datasets/tyqiangz/multilingual-sentiments

Evaluation

Criterion	Weight
Implementation	35%
Cross-lingual experimentation	25%
Model comparison	20%
Report quality & discussion	10%
Presentation	10%

Deliverables (in both cases)

Students must submit:

- A code repository containing the implementation and instructions for reproducing results, including also a dataset description.
- A technical report (8–10 pages) describing methodology, experiments, and analysis.

Technical report (in both cases)

A scientific article based on **Springer Conference Proceedings** template should be provided. The template that can be found here and feel free to use Word or LaTex based on your preferences: https://www.springer.com/gp/computer-science/lncs/conference-proceedings-guidelines?srsltid=AfmBOoplA1uGeD6GeFfyAVWOqk-nUXnBuArVuJnH3PLzdJ3l-Zi_zqy1

Number of pages: 8-10pages (max). Any additional page **will lead to 1 point deduction** out of the total 10points assigned to he Technical Report quality (presented before).